

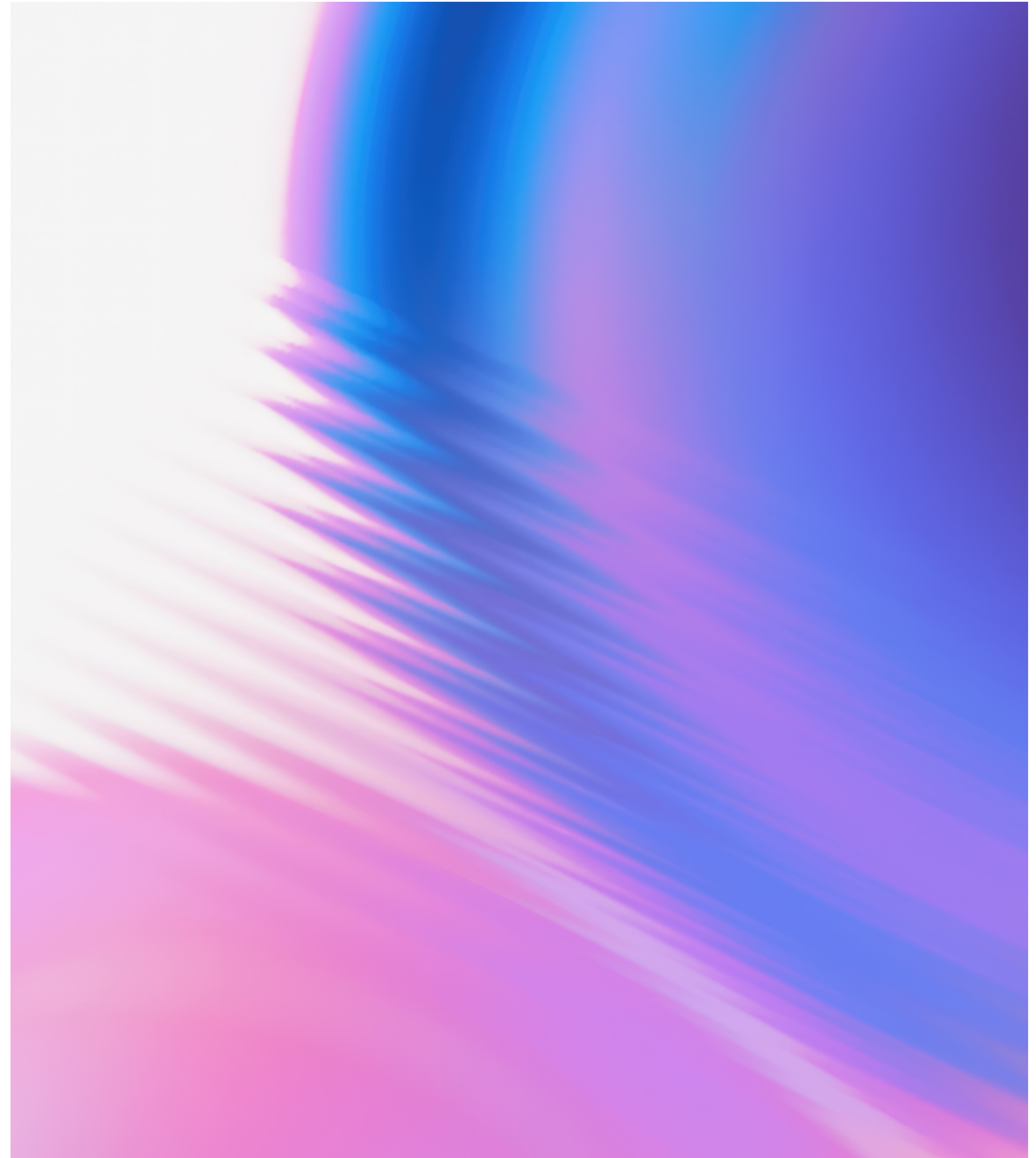
Deploying fermionic simulations on quantum-centric supercomputers

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Agenda

`ibm.biz/qce26-tut-297`



Session 1 (10:00–11:30 AM)

- Review quantum simulation of fermions
- Construct and simulate quantum circuits with `ffsim`
- Approximate ground state energies with SQD and SKQD
- Coding session

Session 2 (1:00–2:30 PM)

- Quantum resource management with QRMI
- Deploy fermionic simulations within a quantum-centric supercomputing workflow

Quantum simulation of fermions

Simulating fermions is one of the most anticipated applications of quantum computers, with applications in **catalysis**, **materials design**, and **drug discovery**.

The starting point for fermionic simulations is the **electronic structure problem**.

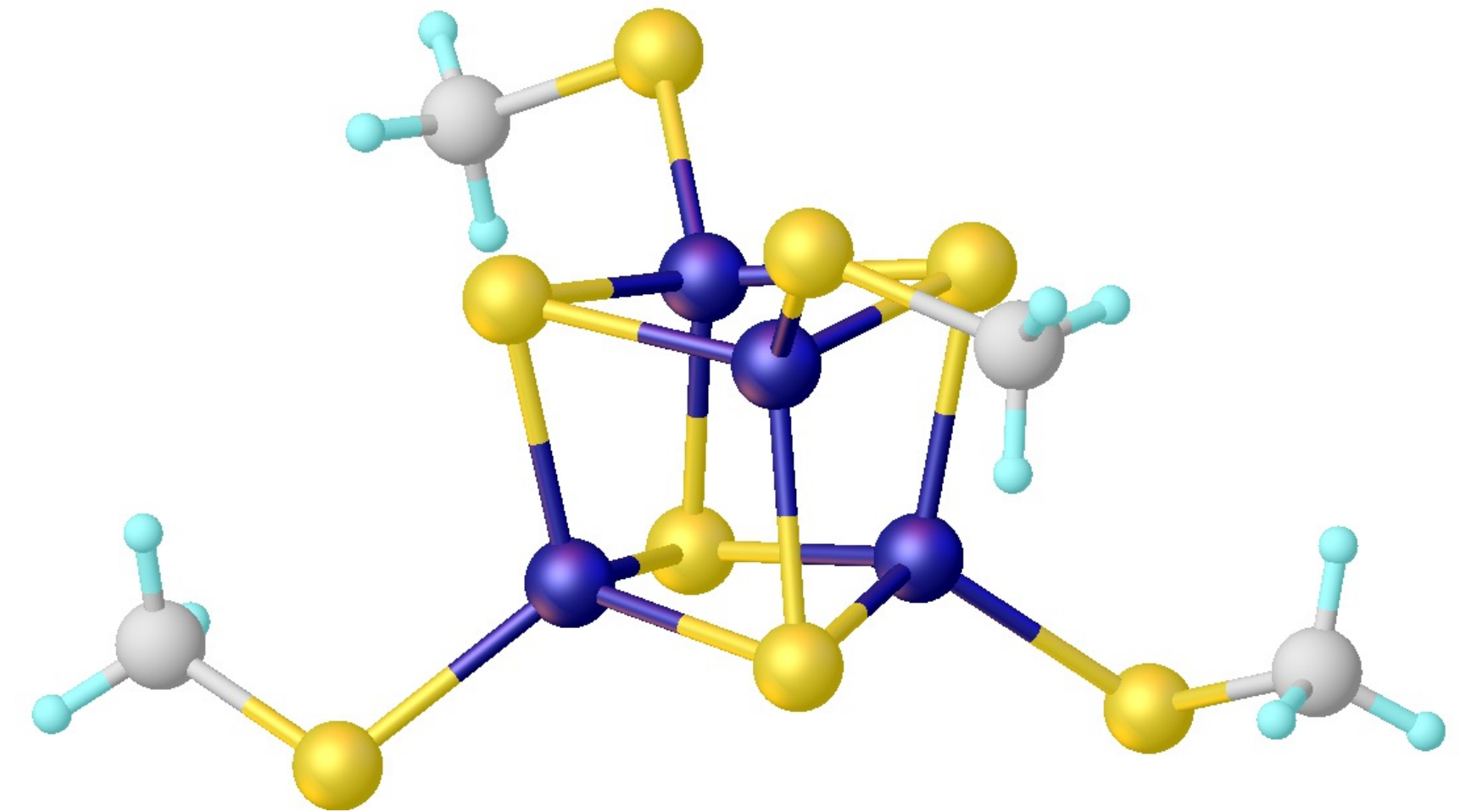


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The electronic structure problem

Electronic structure Hamiltonian:

$$H = \sum_{\substack{pq \\ \sigma}} h_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \frac{1}{2} \sum_{\substack{pqrs \\ \sigma\tau}} h_{pqrs} a_{p\sigma}^\dagger a_{r\tau}^\dagger a_{s\tau} a_{q\sigma}$$

$p, q \in \{1, \dots, N\}$

$\sigma, \tau \in \{\uparrow, \downarrow\}$

N : number of orbitals

$a_{p\sigma}$: fermionic annihilation operator
($2^{2N} \times 2^{2N}$ matrix)

Goal:

Find lowest eigenvalue and corresponding eigenvector (ground state and ground state energy) of H .

Computer representation:

h_{pq} : $N \times N$ matrix

h_{pqrs} : $N \times N \times N \times N$ tensor

Common file format: FCIDUMP

Selected configuration interaction (SCI) methods

Schrodinger equation:

$$H|\psi\rangle = E|\psi\rangle$$

This is an eigenvalue equation in an exponentially large vector space.

Selected configuration interaction (SCI) family of methods:

Replace

$$H \rightarrow H_S = P_S H P_S$$

where

$$P_S = \sum_{x \in S} |x\rangle \langle x|$$

P_S is a projector onto the subspace spanned by the basis vectors in the set S .

H_S approximates H with dimension $|S|$. It can be solved on a classical computer.

Selected configuration interaction (SCI) methods

When is H_S an acceptable approximation to H ?

S must contain the configurations (bitstrings) necessary to represent the ground state.

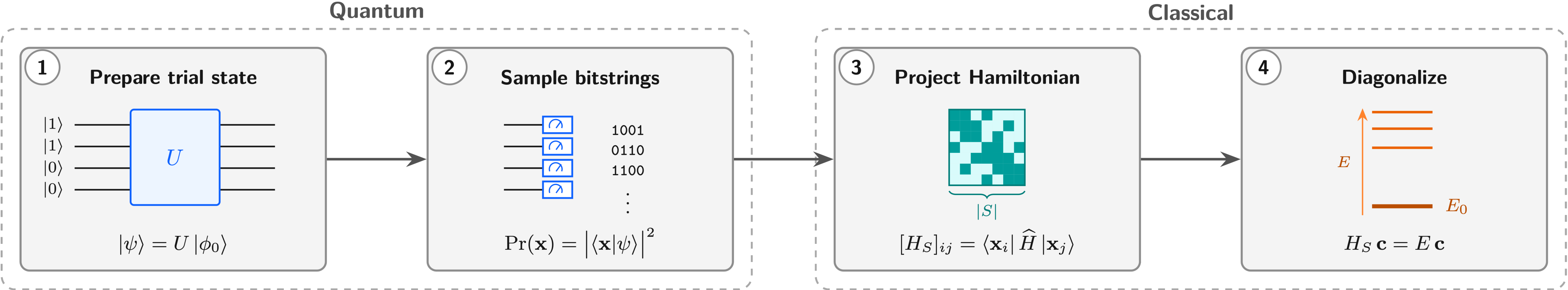
Classical heuristic for picking S :

Semistochastic heat-bath configuration interaction (SHCI), implemented by the Dice software.

Quantum heuristic for picking S :

Quantum-selected configuration interaction (QSCI).

Quantum-selected configuration interaction (QSCI)



Issue: On noisy QPUs, sampled bitstrings have errors!

Symmetries

Example:

norb = 6

nelec = (3, 2)

Hartree-Fock configuration:

000011000111
spin down spin up

Valid configurations:

000101011001

100010010110

Invalid configurations:

000111011001

(wrong number of spin down electrons)

000111010010

(wrong number of spin up and spin down electrons, despite correct total electron count)

Symmetries

Example:

`norb = 6`

`nelec = (3, 2)`

Hartree-Fock configuration:

000011000111
spin down spin up

The dimension of the vector space spanned by valid configurations is

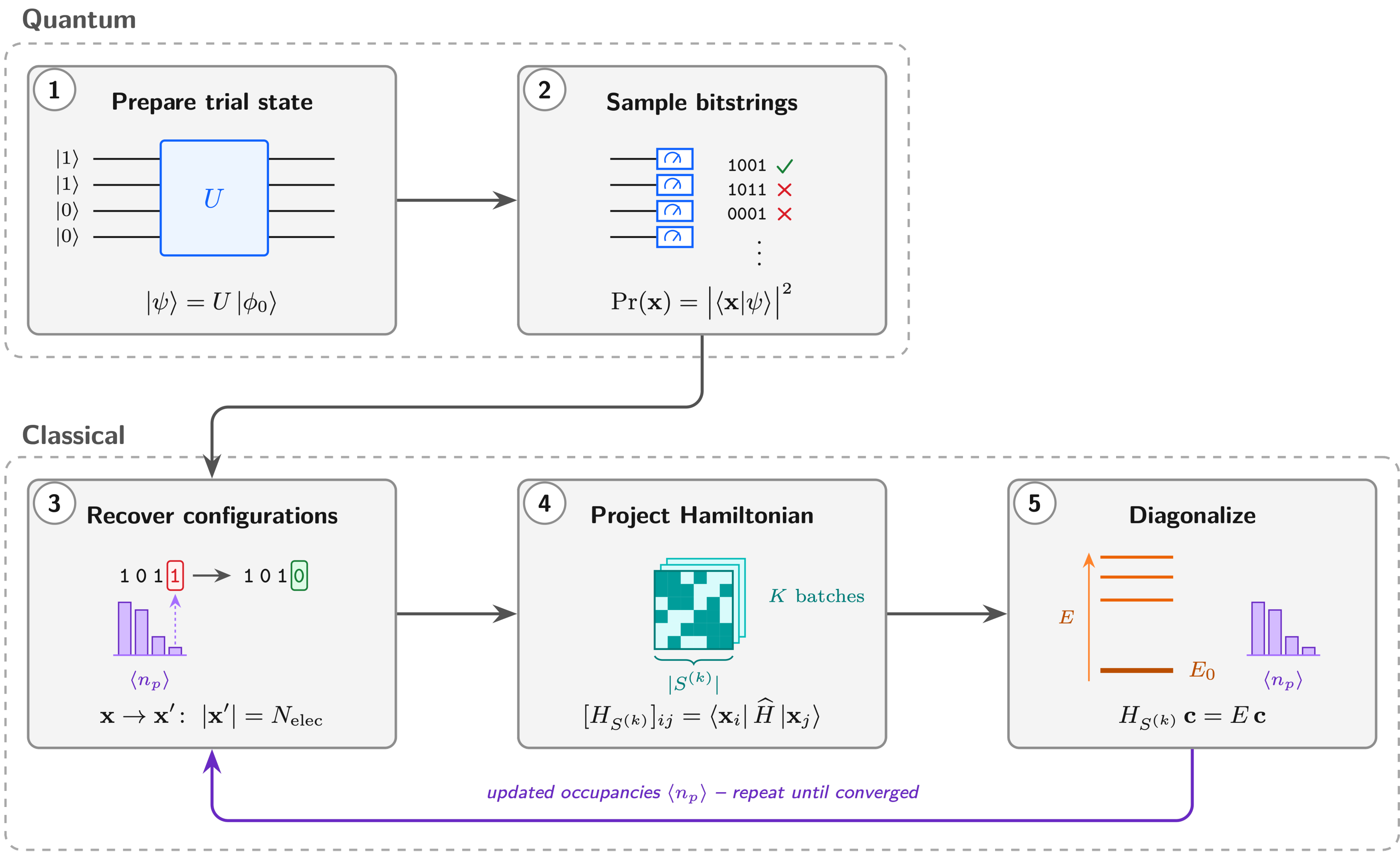
$$\binom{N}{N_{\alpha}} \times \binom{N}{N_{\beta}}$$

By comparison, the total number of possible bitstrings of length $2N$ is

$$2^{2N}$$

`ffsim` performs state vector simulation in the reduced symmetry subspace.

Sample-based quantum diagonalization (SQD)



`ffsim`

Build and simulate Qiskit circuits for the LUCJ ansatz.

`qiskit-addon-sqd`

Run the SQD configuration recovery loop.

`fulqrum`

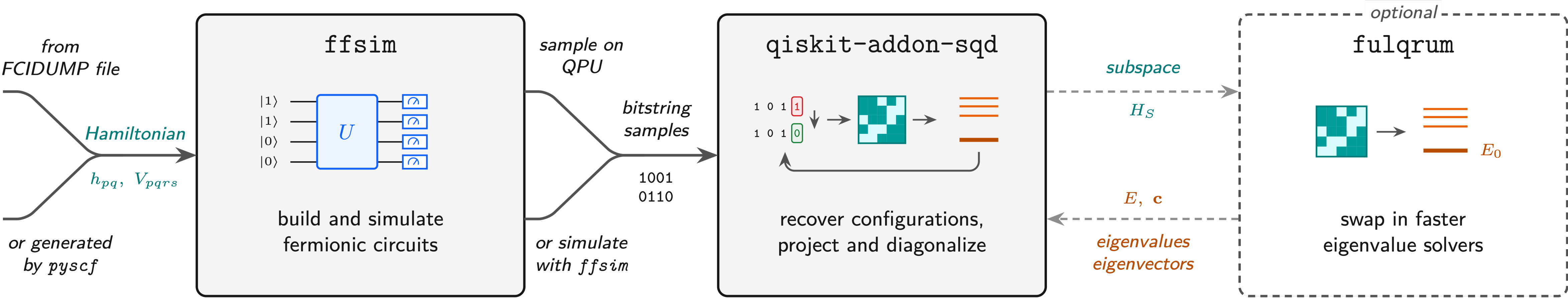
Interface to faster eigenvalue solvers for SQD.

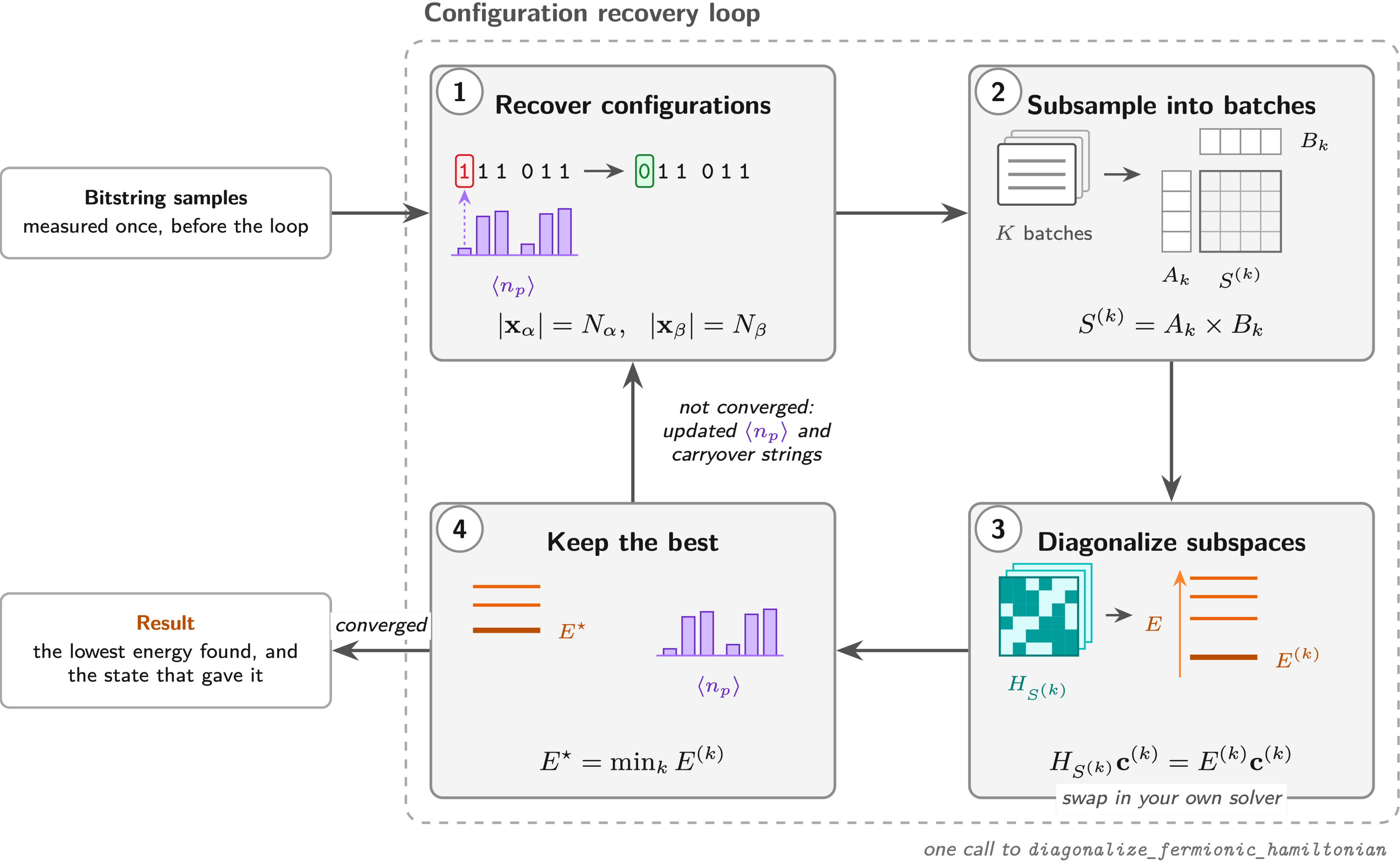
`qrmi`

Vendor-agnostic library to manage QPU resources on HPC systems.

`qiskit-fermions`

Fermionic circuit class, fermion-to-qubit mappings, and circuit transpilation beyond Jordan-Wigner.





Fast **state vector simulation** of fermionic circuits, exploiting particle number and spin z symmetry

Construct quantum circuits for **variational ansatzes** and **Hamiltonian time evolution**

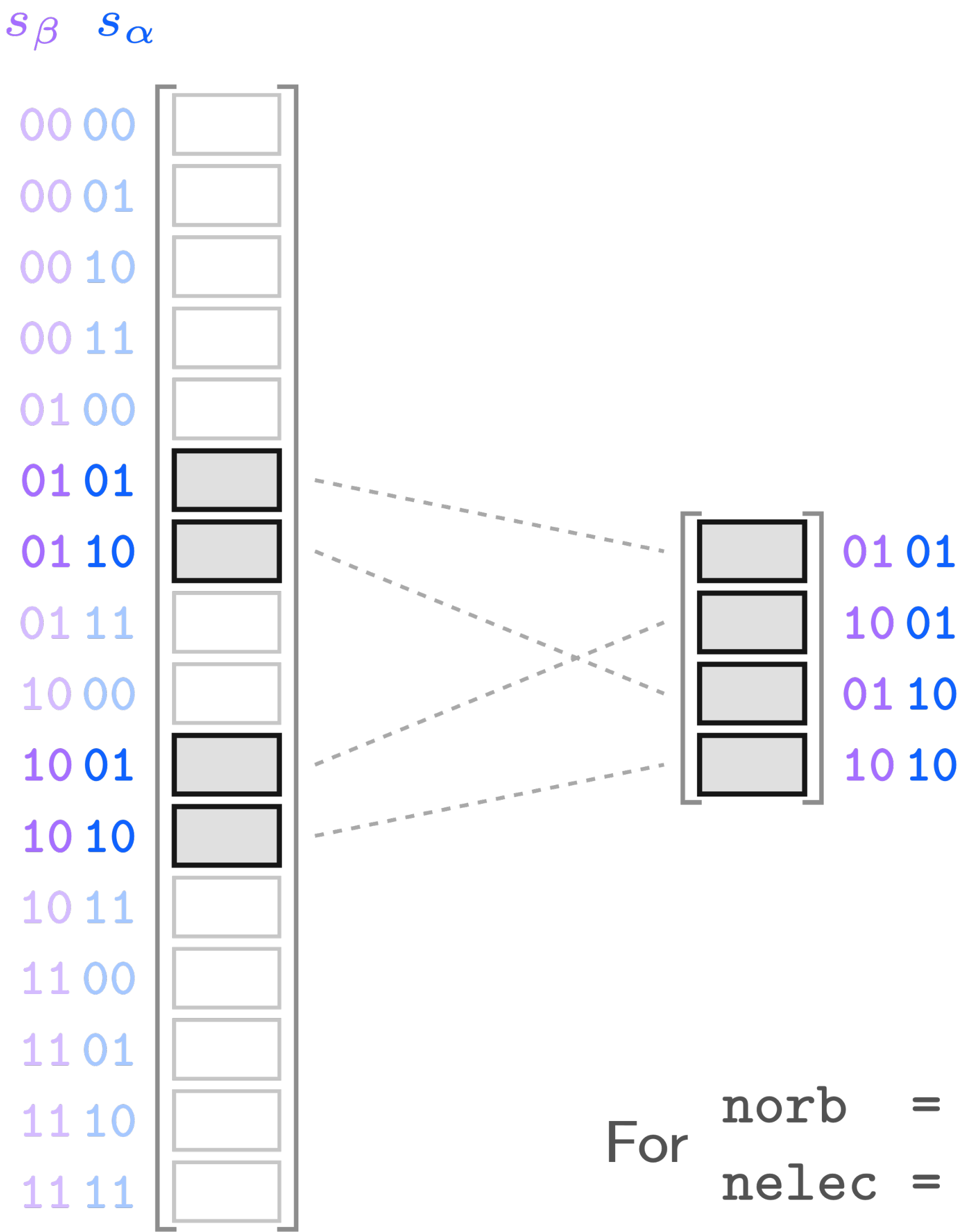
Integrates with **Qiskit** and **PySCF**

Additional features like **compressed double factorization** and **efficient sampling of Slater determinants**, as well as **comprehensive documentation**

Example: `norb = 2`
`nelec = (1, 1)`

a generic simulator's
state vector:

$2^{2N} = 16$ amplitudes



ffsim's state vector:
 $\binom{N}{N_\alpha} \binom{N}{N_\beta} = 4$ amplitudes

For `norb = 20`
`nelec = (5, 5)` : 16 TiB → 3.6 GiB

Coding session notebooks
(links in agenda website)

Sample-based quantum diagonalization of a
chemistry Hamiltonian

`ibm.biz/sqd-tutorial`



Try the following:

- Use `fulcrum` to substitute a different eigenvalue solver such as Scipy or PRIMME. Does it get faster?
- Change the small-scale basis from STO-6G to 6-31G. How many orbitals are there now? Can it be simulated with `StatevectorSampler`? What about `ffsim.FfsimSampler`?
- Edit the following hyperparameters:
 - `max_iterations`
 - `num_batches`
 - `samples_per_batch`
 - `symmetrize_spin`
 - `max_dim`
 - `carryover_threshold`Try to find a setting that favorably balances cost and accuracy.